FIN2020 Homework 7

Problem 1

Assume the following market model:

$$\tilde{r}_j = \alpha_j + \beta_{jM}\tilde{r}_M + \tilde{\varepsilon}_j$$

with

$\tilde{r}_j =$ the rate of return on asset j
$\alpha_j =$ a constant
$\beta_{jM} =$ the sensitivity of asset j to fluctuations in the market return
$\tilde{r}_M =$ the market rate of return
$\varepsilon_j =$ a stochastic component with $E(\tilde{\varepsilon}_j) = 0$, $Var(\tilde{\varepsilon}_j) = \sigma^2_{\varepsilon_j}$ and $cov(\tilde{\varepsilon}_i, \tilde{\varepsilon}_j) = 0, \forall i, j$.

a. Prove that $\sigma^2_j = \beta^2_j \sigma^2_M + \sigma^2_{\varepsilon j}$.
b. Prove that $\sigma_{ij} = \beta_i \beta_j \sigma^2_M$.

Problem 2

Consider an economy in which the random return $\tilde{r}_i$ on each individual asset i is given by

$$\tilde{r}_i = E(\tilde{r}_i) + \beta_{i,m}[\tilde{r}_M - E(\tilde{r}_M)] + \beta_{i,v}[\tilde{r}_V - E(\tilde{r}_V)] + \varepsilon_i$$

where, as we discussed in class, $E(\tilde{r}_i)$ equals the expected return on asset i, $\tilde{r}_M$ is the random return on the market portfolio, $\tilde{r}_V$ is the random return on a "value" portfolio that takes a long position in shares of stock issued by smaller, overlooked companies or companies with high book-to-market values and a corresponding short position is shares of stock issued by larger, more popular companies or companies with low book-to-market values, ε_i is an idiosyncratic, firm-specific component, and $\beta_{i,m}$ and $\beta_{i,v}$ are the "factor loadings" that measure the extent to which the return on asset i is correlated with the return on the market and value portfolios. Assume, as Stephen Ross did when developing the arbitrage pricing theory (APT), that there are enough individual assets for investors to form many well-diversified portfolios and that investors act to eliminate all arbitrage opportunities that may arise across all well-diversified portfolios.

a. Consider, first, a well-diversified portfolio that has $\beta_{w,m} = \beta_{w,v} = 0$. Write down the equation, implied by the APT, that links the expected return $E(\tilde{r}_w^1)$ on this portfolio to the return r_f on a portfolio of risk-free assets.

b. Consider, next, two more well-diversified portfolios. portfolio two with $\beta_{w,m} = 1$ and $\beta_{w,v} = 0$ and portfolio three with $\beta_{w,m} = 0$ and $\beta_{w,v} = 1$. Write down the equations, implied by this version of the APT, that link the expected returns $E(\tilde{r}_w^2)$ and $E(\tilde{r}_w^3)$ on each of these two portfolios to r_f, $E(\tilde{r}_M)$, and $E(\tilde{r}_V)$.

c. Suppose you find a fourth well-diversified portfolio that has non-zero values of both $\beta_{w,m}$ and $\beta_{w,v}$ and that has expected return

$$E(\tilde{r}_w^4) = r_f + \beta_{w,m}[E(\tilde{r}_M) - r_f] + \beta_{w,v}[E(\tilde{r}_V) - r_f] + \Delta,$$

where $\Delta < 0$ is a negative number. Explain briefly how you could use this portfolio, together with the first three from parts (a) and (b), above, in a trading strategy that involves no risk, requires no money down, but yields a future profit for sure.

Problem 3

CAPM and APT:

a. Briefly describe these two models: Capital Asset Pricing Model and Arbitrage Pricing Theory? What are their main assumptions?

b. CAPM and APT are so-called valuation models. In fact, they permit us to compute the *expected rate of return* of a security. How can this be used to derive a *current price*?

c. Contrast the two models. Are they compatible? Are they identical under certain conditions?